Ivan Ricardo

PhD Candidate in Econometrics
School of Business and Economics, Maastricht University

iu.ricardo@maastrichtuniversity.nl
ivanuricardo.github.io
github.com/ivanuricardo
Citizenship: USA

EDUCATION

2023 – present **PhD in Econometrics** — Maastricht, NL
Maastricht University · Expected completion: 2027
Thesis: *Multilinear Time Series Econometrics*
Advisors: Alain Hecq and Ines Wilms

Fall 2026 **Visiting PhD Researcher** — Carbondale, IL
Southern Illinois University · Host: S. Yaser Samadi

2021 – 2023 **MSc in Economic and Financial Research** — Maastricht, NL
Maastricht University · Specialization in Econometrics

2017 – 2020 **BA in Economics** — Santa Barbara, CA
University of California, Santa Barbara

RESEARCH INTERESTS

Time series econometrics, matrix- and tensor-valued time series, dimension reduction, impulse response analysis

JOB MARKET PAPER

Impulse Response Inference for Matrix Autoregressions ARXIV
with A. Hecq and I. Wilms
Abstract: Matrix autoregressive (MAR) models offer a parsimonious framework for matrix-valued time series, yet tools for quantifying uncertainty around their impulse responses are lacking. We develop asymptotic and bootstrap-based inference for impulse responses of stable MAR(p) models. We derive the joint asymptotic distribution of the coefficient and covariance estimators, which yields closed-form delta-method standard errors. To address finite-sample bias, we propose ProBAB-MAR, a bias-corrected bootstrap that projects the corrected coefficients back onto the Kronecker parameter space, and prove that it attains asymptotically correct coverage. Monte Carlo simulations show that delta-method intervals undercover in small samples, while ProBAB-MAR achieves near-nominal coverage with intervals considerably narrower than those from an unrestricted VAR. An application to Euro-area inflation illustrates how shocks transmit across countries and inflation categories.

PUBLICATIONS

2026 **Decomposing Co-movements in Matrix-Valued Time Series: A Pseudo-Structural Reduced Rank Approach**
A. Hecq, **I. Ricardo**, and I. Wilms
Econometrics and Statistics JOURNAL

2026 **Performance of Radio-Based Detection for Operational Monitoring of M5+ Class Solar Flares**
J. van Ravenswaaij, I. Wilms, **I. Ricardo**, and M. Brchnelova
Journal of Space Weather and Space Climate JOURNAL

2025 **Detecting Cointegrating Relations in Non-Stationary Matrix-Valued Time Series**
A. Hecq, **I. Ricardo**, and I. Wilms
Economics Letters, 248, 112205 JOURNAL

WORK IN PROGRESS

Structured Dimension Reduction for High Dimensional VAR Models
with S. Yaser Samadi

CONFERENCES, WORKSHOPS, AND SEMINARS

Presentations

2026 Netherlands Econometrics Study Group (poster), Midwest Econometrics Group, Southern Illinois University Colloquium*, CFE-CMStatistics* (scheduled)

2025 CFE-CMStatistics*

2024 CFE-CMStatistics*, Econometrics Seminar (Maastricht University), Netherlands Econometrics Study Group (poster)

* Invited

Participation

2023 Network Econometrics Summer School, Ca' Foscari University of Venice

2022 Dimensionality Reduction and Inference in High-Dimensional Time Series

TEACHING

Tutor — Maastricht University — Avg. evaluation

Econometric Methods I (BSc. Econometrics & Operations Research) 8.60/10
OLS, failure of Gauss–Markov assumptions (omitted variables, heteroskedasticity, and autocorrelation), misspecification tests, unit roots, and cointegration.

Macroeconomics (BSc. Econometrics & Operations Research) 8.69/10
Modern macroeconomic theory with a focus on short run fluctuations, medium run equilibrium, and long-run growth, with applications to global economic policy.

Quantitative Methods III (BSc International Business) 8.35/10
Applies inferential statistics tools including t-tests, ANOVA, chi-square tests, and regression analysis for business research.

Statistics I (BSc. University College Maastricht) 4.7/5
Descriptive and inferential statistics for the social and life sciences, from confidence intervals to hypothesis testing, with data analyses in JASP.

	Applied Statistics (BSc. Maastricht Science Programme) Intermediate statistics for sciences, from probability distributions and sampling theory to confidence intervals, hypothesis testing, regression, and ANOVA.	4.5/5
Guest lecturer	Econometric Methods I (Maastricht University) Introduction to the Julia Programming Language	2024, 2025
	Time Series (Undergraduate/graduate, Southern Illinois University) Impulse Response Inference in Time Series Econometrics	2026
Supervision	Thesis supervision (Econometrics & OR) Topics include graphical models for matrix time series, autoregressive models for matrix time series, time series clustering, and solar flare forecasting	1 MSc, 4 BSc

SOFTWARE

Julia — Creator and maintainer of the following open-source packages

CommonFeatures.jl **GraphicalLasso.jl**
MatrixAutoRegressions.jl **PseudoStructuralComovements.jl**

SKILLS

Programming	Julia, R, Python, MATLAB, Bash, Lua
Tools	Git, Docker, Linux, LaTeX
Languages	English and Spanish (native), Dutch (basic)

REFERENCES

Alain Hecq
Professor
Maastricht University
a.hecq@maastrichtuniversity.nl

Ines Wilms
Associate Professor
Maastricht University
i.wilms@maastrichtuniversity.nl

S. Yaser Samadi
Associate Professor
Southern Illinois University
ysamadi@siu.edu